

AITentive: A Toolkit to Develop Reinforcement Learning-based Attention Management Systems

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ABSTRACT

In today's fast-paced world, multitasking is common and affects productivity, decision-making, and cognition. Understanding its complexities is crucial for improving well-being, efficiency, and task management. Attention management systems optimize notification and interruption timings. This work introduces *AITentive*, an open-source Unity3D toolkit for multitasking research and developing attention management systems with reinforcement learning. The toolkit offers customizable tasks, built-in measurements, and a uniform interface for adding tasks, using Unity ML agents to develop and train attention management systems based on user models.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models; **Interactive systems and tools**; *Ubiquitous computing*; **User interface management systems**.

KEYWORDS

Tasks/Interruptions/Notification, Programming/Development Support, Information Seeking & Search, Machine Learning

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1 INTRODUCTION

Imagine writing an important document while responding to short text messages, or driving and selecting navigation alternatives due to congestion. Both scenarios involve deciding when to switch attention from a primary task (writing or driving) to a secondary one (responding to messages or using navigation). Research indicates humans struggle with such switches, especially when externally interrupted, leading to decreased performance, increased stress, and errors [4, 8, 11]. Effects range from workplace performance drops

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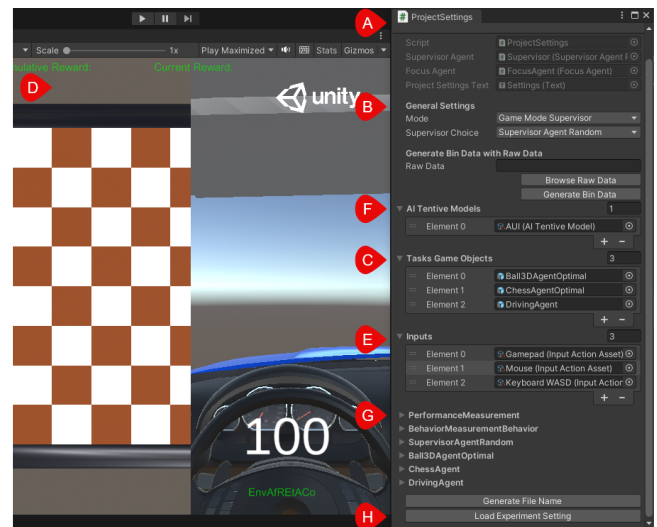


Figure 1: Centralized component to configure the project.

to severe traffic accidents [1, 9, 15]. Today, most services notify users immediately, disregarding optimal interruption times [19]. To counter this, so-called “Attention Management Systems” (AMSs) aim to consider the impact of interruptions on user attention and time communication contextually [2, 5, 7], with the ultimate goal to minimize disruption while delivering information efficiently [3, 4]. Developing AMSs is challenging, often relying on rigid computational theories or complex, data-intensive approaches [3, 13]. We have developed a novel approach, which uses reinforcement learning (RL) to optimize interruption timing and enhance multitasking performance [10].

In this work, we present an extension of this architecture and offer a publicly available toolkit¹ for researching RL-based AMSs across scenarios like driver-vehicle interaction [18], manufacturing [6, 14], human-robot interaction, workplaces [12], or VR [17].

2 DEVELOPING AMS WITH RL

RL operates without labeled training data, defining an agent through environmental states, actions, and a reward function. Actions lead the agent through different states, with rewards or penalties based on the quality of successor states [16]. In RL-based AMSs, the agent

¹<https://github.com/reapphil/AITentiveMultitasking>

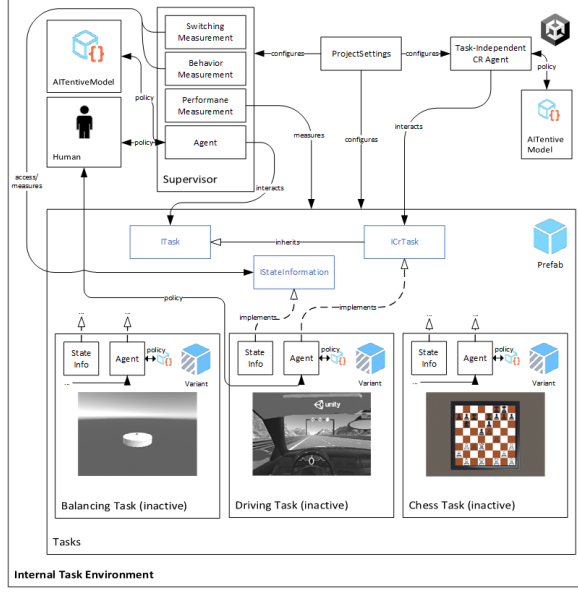


Figure 2: Component relationships within the toolkit.

communicates task interruptions/switches to the user by observing interactions and task dynamics guided by learned policies. Recently, we have shown that such a system can support humans in reaching “superhuman” multitasking performance [10].

However, designing, implementing, training, and evaluating RL-based AMSs is quite complex, as this requires implementing (1) the AMS, (2) tasks, (3) measurement components, (4) defining their interfaces, (5) building an user simulation for the training of the AMS, (6) data collection of humans performing the task, (7) training of task agents, (8) data collection of agents performing the task, (9) evaluation of the similarity between human and agents, (10) training of the AMS with the best fitting task agent, and (11) conducting a user study to investigate its potential.

Despite the potential of the general concept, researchers may see barriers in this complex process, hindering them from engaging in this promising field. In response, we developed our toolkit, *AITentive*, to solve steps 1, 3, and 4, while also offering functions and standardized interfaces to support the resolution of the remaining steps. The remainder of this work describes the architecture of the toolkit and how the mentioned steps are addressed.

3 THE TOOLKIT

The toolkit utilizes the Unity game engine as a simulation environment and provides the ability to define custom tasks and agents (e.g. humans or RL-agents) by implementing the necessary interfaces (see 4). The tasks can be integrated into an overarching RL-based AMS that decides which tasks should be attended so that a configurable reward function increases.

Figure 1 displays the Unity editor with the *ProjectSettings* component (A) for configuring the simulation environment. The *General Settings* (B) determine whether task switching is controlled by the AMS or the user, with AMS switches set to forced, notified, or suggested. Tasks that meet the necessary interfaces can be added to

the *Tasks Game Objects* list (C) and will appear in the game view (D). The *AITentive Models* list (F) holds the RL models for the task- and AMS-agents, automatically assigning them to the appropriate agent. Section (E) assigns predefined control configurations to each task. Section (G) includes menus for configuring tasks, the AMS, and measurement components. Section (H) allows the import of predefined experiment configurations in JSON format (supporting 11).

4 SOLUTION

Figure 2 shows the internal task environment’s architecture. Tasks extend a *Prefab* including mandatory components like a camera and an *ITask* interface (supporting 2) for AMS agent interaction (solving 4). The already implemented AMS of the toolkit controls the currently active task, indicated by a colored output, while inactive tasks are in black and white (solving 1). The controlling entity uses an agent script from the *ML-Agents* framework (supporting 5), operating with either a learned model or user input. The *ICrTask* interface defines methods for any agent within the CR environment (e.g., a vision agent). Tasks and components are configured through *ProjectSettings*, either directly in Unity or via configuration files.

The toolkit includes measurement features to estimate performance, switching behavior, and task behavior (solving 3). Performance, defined by the AMS’s reward function, is task-independent. Switching and task behavior measurements are task-dependent but standardized. The *IStateInformation* defines the state space per task, while the *Behavior Measurement* component records average actions in a discretized space for comparison. Reaction time uses discretization based on source and target tasks. Switching behavior is recorded per task combination, capturing the state space and actions at switching moments.

The toolkit also provides pipelines to train (supporting 7 and 10) and evaluate the involved agents (supporting 6 and 8). The training and evaluation pipelines create player instances based on provided configuration files. For training, these instances use the *ML-Agents* framework, allowing concurrent training across multiple instances, or external algorithms can be used. The evaluation environment tests different models against targets, like human behavior. Once set up, the pipeline executes evaluations, measuring objectives (e.g., reaction times, average actions) for a predefined number of actions. This data determines the distance between model and target behavior using the toolkit’s distance library (supporting 9).

5 CONCLUSION

Our *AITentive* toolkit enables research into complex RL-based AMS systems with a low entry barrier. We are using it to showcase recent advances [10] in real-life scenarios like multitasking in vehicles, air traffic control, and factory automation. As devices and services proliferate, systems that account for users’ strengths and limitations in timing interruptions and delivering information will become even more important in the future.

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